

Structural properties of one shell-generated C^* -algebra

MathlibAnnex.CStarAlgebra.CAR.shellFamilyEndpoint

theorem

Collects the proved properties of a single generated algebra, keeping the shell-family input separate from its consequences.

Statement

For every fixed shell family, form T , ι and ρ_0 as below. Then T is nonzero, norm closed and infinite-dimensional, ι is injective and unital, and ρ_0 is faithful and irreducible. Every nonzero irreducible complex star representation of T , even if not initially required to preserve the unit, is unitarily equivalent to ρ_0 . The only norm-closed two-sided ideals of T are 0 and T , and T has no injective star representation whose range is precisely all compact operators on any complex Hilbert space.

Assumptions

Let A be the completed CAR algebra with matrix stages $M_{2^n}(\mathbb{C})$ embedded by $a \mapsto a \otimes I_2$. Write f_n for the image of the first diagonal matrix unit, $p_n = f_n - f_{n+1}$, and ω for the root state characterized by $\omega(\iota_n(a)) = a_{00}$ on each stage, where ι_n is its canonical embedding.

Index chosen pure-state Gelfand-Naimark-Segal (GNS) representations by their unitary-equivalence classes $j \in I$, with root index o and representative states φ_j .

A fixed shell family consists of complex-linear unital star automorphisms α_j and elements $w_{j,n} \in A$ satisfying $\varphi_j(\alpha_j(a)) = \omega(a)$, $w_{j,n}^* w_{j,n} = \alpha_j(p_n)$ and $w_{j,n} w_{j,n}^* = p_n$.

At the root, α_o is the identity and $w_{o,n} = p_n$.

Let $\mathcal{H} = \bigoplus_{j \in I} H_j$ be the Hilbert direct sum of these GNS spaces, $\pi = \bigoplus_j \pi_j$ their representation of A , and $b_j = e_j \xi_j \in \mathcal{H}$ its embedded unit cyclic vector, where $\xi_j \in H_j$ is the selected unit cyclic vector and $e_j : H_j \rightarrow \mathcal{H}$ is coordinate inclusion.

Choose once a family of unitary operators $L_j \in B(\mathcal{H})$ from the construction of the shell unitaries: $L_j b_j = b_o$, $L_j \pi(\alpha_j(p_n)) = \pi(w_{j,n})$, and $L_o = I_{\mathcal{H}}$. Here $B(\mathcal{H})$ denotes the bounded complex-linear operators on $\mathcal{H}$.

Put $T = C^*(\pi(A), \{L_j : j \in I\}) \subseteq B(\mathcal{H})$, the norm-closed unital star algebra generated by these operators. Let $\iota : A \rightarrow T$ be $\iota(a) = \pi(a)$ with its codomain restricted to T , and let $\rho_0 : T \rightarrow B(\mathcal{H})$ be inclusion.

The comparison Hilbert space K may be any complete complex Hilbert space in an arbitrary independent universe. No separability or dimension bound on K is added. The chosen family, links and algebra T do not depend on that comparison universe.

Conclusion

All conclusions concern this same concrete T . In the capture assertion, unitary equivalence means there is a surjective complex-linear isometry $U : \mathcal{H} \rightarrow K$ with $U(\rho_0(t)x) = \rho(t)(Ux)$ for all $t \in T$ and $x \in \mathcal{H}$. In the compact-operator assertion, the competing map need not be unital and the zero Hilbert space is also covered.

The separately linked Lean structure lists these properties; defining that proposition does not prove it. This theorem proves that proposition for every supplied family. The family hypothesis remains explicit. The structural theorem for the homogeneity-based algebra and its faithful representation on a separable Hilbert space specialize this fixed-family result.

Proof route

The construction of the shell unitaries gives a faithful source map and irreducible inclusion. The unital copy of the infinite-dimensional CAR algebra makes T nonzero and infinite-dimensional. The universal capture theorem supplies unitary equivalence for unital irreducible representations; irreducibility forces any nonzero possibly nonunital representation to preserve the unit. Ideal-separating pure GNS representations then yield simplicity. Finally a unital infinite-dimensional algebra cannot be represented injectively onto all compact operators.

▼ Proof steps

1. Choose the links once, restrict π to the generated algebra, and use its faithful root summand. Norm closure is part of the construction of T .
2. If T were finite-dimensional, its injective complex-linear source map would make $\mathcal{A}$ finite-dimensional, contradicting the growing matrix stages.
3. Apply the unitary-equivalence theorem, the nonunital-to-unital argument, and the closed-ideal consequence. Exclude an isomorphism of this same infinite-dimensional algebra onto all compact operators. These are separate proved results supplying the listed fields.

Main citations

- Exact statement and proof
- Properties of the fixed target
- A fixed family of compatible shell data
- Existence of the completed shell model
- Choosing one system of unitary links
- The chosen closed generated algebra
- The source map into the generated target
- Inclusion into the ambient operators
- Infinite dimension of the generated target
- Every irreducible representation is equivalent to the faithful inclusion
- Unitary capture without assumed unitality
- Closed-ideal simplicity of the target
- No model onto all compact operators

Lean source signature (exact)

```
theorem shellFamilyEndpoint (family : RepresentativeShellFamily) :
  ShellFamilyEndpoint.{v} family
```

In the source	Mathematical meaning
family : RepresentativeShellFamily	The supplied normalized CAR shell family fixes the unitary links L_j , the atomic space $\mathcal{H}$, the generated algebra T , the source $\iota : A \rightarrow T$, and literal inclusion $\rho_0 : T \rightarrow B(\mathcal{H})$ once for all.
ShellFamilyEndpoint.{v} family	The output is the proposition listing ten properties of those same objects. Its full separate exact structure and one row for every field follow below. The comparison universe v controls the Hilbert-space quantifiers, not a new family or a new target for each field.
theorem shellFamilyEndpoint	This theorem proves all ten fields for every supplied family; merely defining the proposition would not prove them.

The following is a separate, complete exact declaration of the result proposition, from lines 194–229 of the same fixed source. It is not appended to or substituted for the theorem signature above.

```
structure ShellFamilyEndpoint (family : RepresentativeShellFamily) : Prop where
  nontrivial_target : Nontrivial (ShellFamilyTarget family)
  isClosed_target :
    IsClosed
      (ShellFamilyTarget family : Set
        (MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState
→L[ $\mathbb{C}$ ]
          MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert
completedRootPureState))
  source_injective : Function.Injective (shellFamilySourceHom family)
  source_unital : shellFamilySourceHom family 1 = 1
  not_finiteDimensional_target :
    ¬ FiniteDimensional  $\mathbb{C}$  (ShellFamilyTarget family)
  ambient_injective : Function.Injective (shellFamilyInclusion family)
  isIrreducible_ambient :
    Representation.IsIrreducible (shellFamilyInclusion family)
  captures_nonunital :
    ∀ (K : Type v) [NormedAddCommGroup K] [InnerProductSpace  $\mathbb{C}$  K]
      [CompleteSpace K]
      (rho : NonUnitalRepresentation
        (A := ShellFamilyTarget family) (H := K)),
      rho.IsIrreducible →
        ∃ U :
          MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert
completedRootPureState  $\approx_{i_1}$  [ $\mathbb{C}$ ] K,
          ∀ (a : ShellFamilyTarget family)
```

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    (x : MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert
completedRootPureState),
    U (shellFamilyInclusion family a x) = rho a (U x)
closedIdeal_dichotomy :
  ∀ I : TwoSidedIdeal (ShellFamilyTarget family),
    IsClosed (I : Set (ShellFamilyTarget family)) → I = ⊥ ∨ I = T
not_compactOperatorModel :
  ∀ (K : Type v) [NormedAddCommGroup K] [InnerProductSpace ℂ K]
    [CompleteSpace K]
    (e : ShellFamilyTarget family →na [ℂ] (K →L[ℂ] K)),
    ¬ (Function.Injective e ∧
      (∀ a : ShellFamilyTarget family, IsCompactOperator (e a)) ∧
      ∀ T : K →L[ℂ] K, IsCompactOperator T →
        ∃ a : ShellFamilyTarget family, e a = T)

```

In the source	Mathematical meaning
nontrivial_target	The fixed algebra T is nonzero.
isClosed_target	The set $T \subseteq B(\mathcal{H})$ is closed for the operator norm.
source_injective	The fixed source map $\iota : A \rightarrow T$, $\iota(a) = \pi(a)$, is injective.
source_unital	The same map sends 1_A to 1_T .
not_finiteDimensional_target	T is not finite-dimensional over $\mathbb{C}$.
ambient_injective	The literal inclusion $\rho_0 : T \rightarrow B(\mathcal{H})$ is injective.
isIrreducible_ambient	This same ρ_0 is nonzero irreducible: it has no proper nonzero closed reducing subspace.
captures_nonunital	After the family and T have been fixed, for every complete complex Hilbert space K in universe v and every nonzero irreducible possibly nonunital $\rho : T \rightarrow B(K)$, there is a surjective complex-linear isometry $U : \mathcal{H} \rightarrow K$ with $U(\rho_0(t)x) = \rho(t)Ux$ for every $t \in T$, $x \in \mathcal{H}$.
closedIdeal_dichotomy	For every two-sided ideal $\mathfrak{a}$ in T , if its underlying set is norm closed, then $\mathfrak{a} = \{0\}$ or $\mathfrak{a} = T$.
not_compactOperatorModel	For every complete complex Hilbert space K and every possibly nonunital star homomorphism $e : T \rightarrow B(K)$, it is impossible that e is injective, all $e(t)$ are compact, and every compact operator on K equals some $e(t)$. The zero space is included.

Further source notes: Its exact structure declaration is linked as “Properties of the fixed target” above.

▼ **Exact content identity**

Declaration: MathlibAnnex.CStarAlgebra.CAR.shellFamilyEndpoint

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